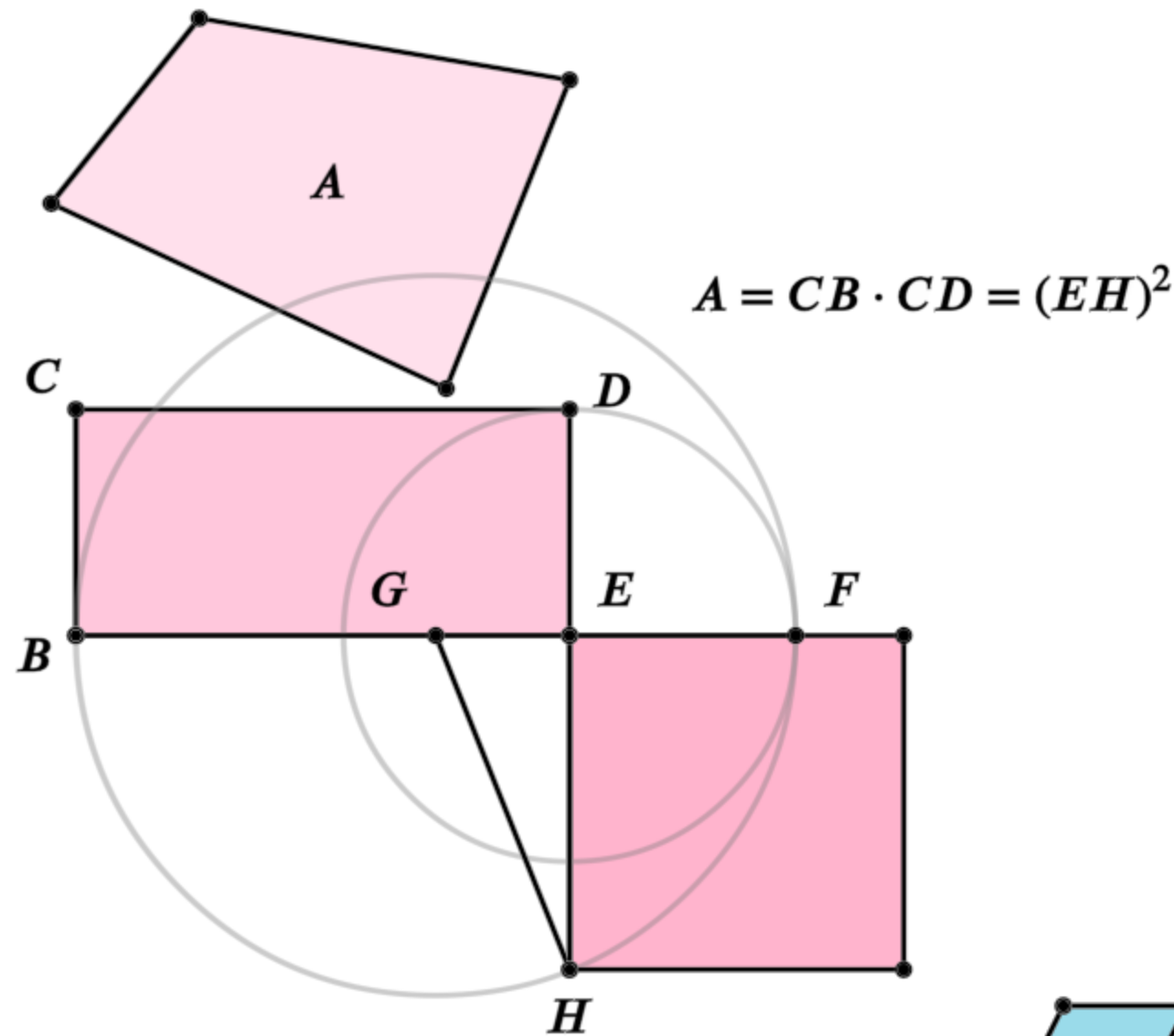


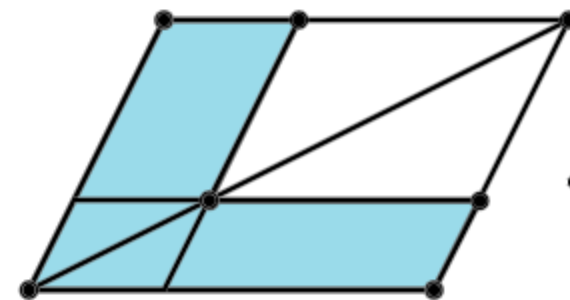
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



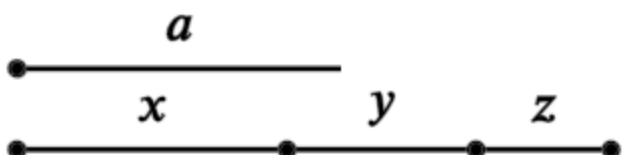
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

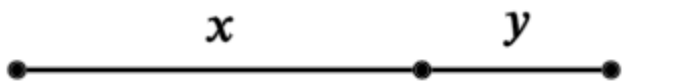
And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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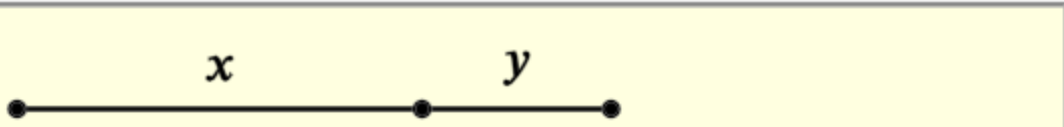
1.


$$a(x + y + z) = ax + ay + az$$

2.


$$(x + y)^2 = x(x + y) + y(x + y)$$

3.


$$y(x + y) = yx + y^2$$

4.


$$(x + y)^2 = x^2 + y^2 + 2xy$$

5.


$$(x + y)(x - y) + y^2 = x^2$$

6.


$$y(2x + y) + x^2 = (x + y)^2$$

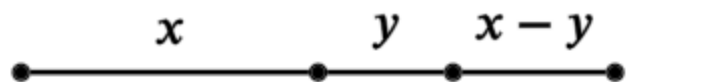
7.


$$(x + y)^2 + y^2 = x^2 + 2y(x + y)$$

8.


$$(x + 2y)^2 = 4y(x + y) + x^2$$

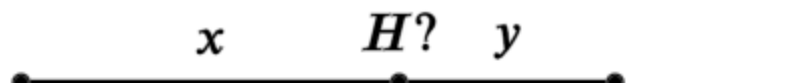
9.


$$(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$$

10.

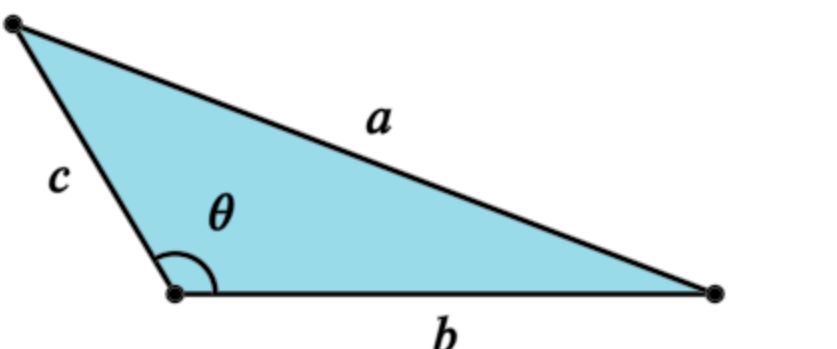

$$(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$$

11.



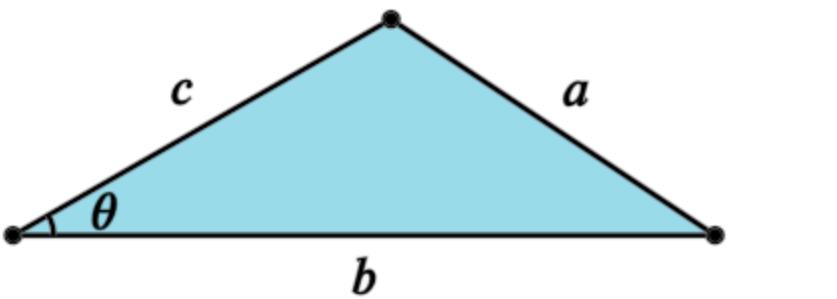
Find H such that: $y(x + y) = x^2$

12.



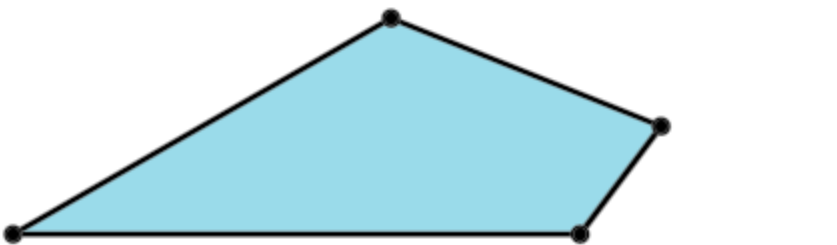
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13.



Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

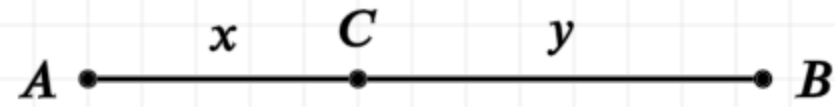
14.



Construct a square equal to the polygon

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

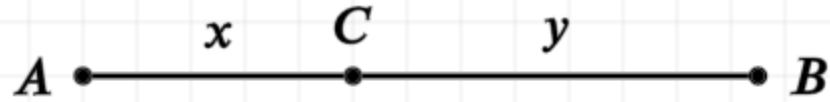
Let AB be a straight line, arbitrarily cut at point C

$$AB = AC + CB$$

$$a = x + y$$

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

Then the area of the rectangle formed by line AB and CB is equal in area to the sum of the rectangles formed by line CB and AC , and line CB and CB

$$AB = AC + CB$$

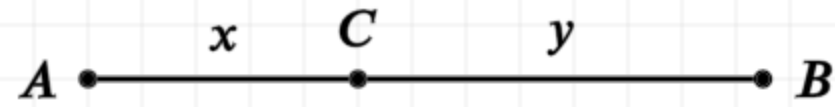
$$CB \cdot AB = CB \cdot AC + CB^2$$

$$a = x + y$$

$$y(x + y) = yx + y^2$$

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

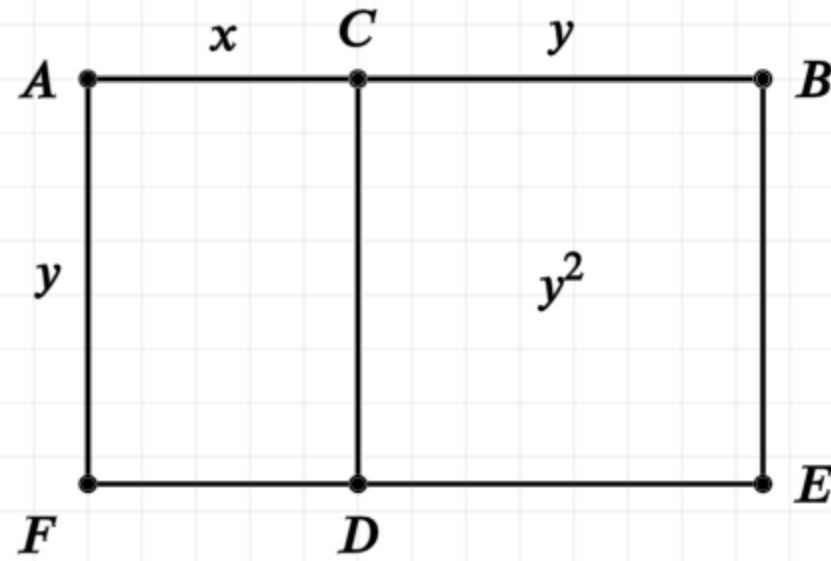
$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

$$AB = AC + CB$$

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

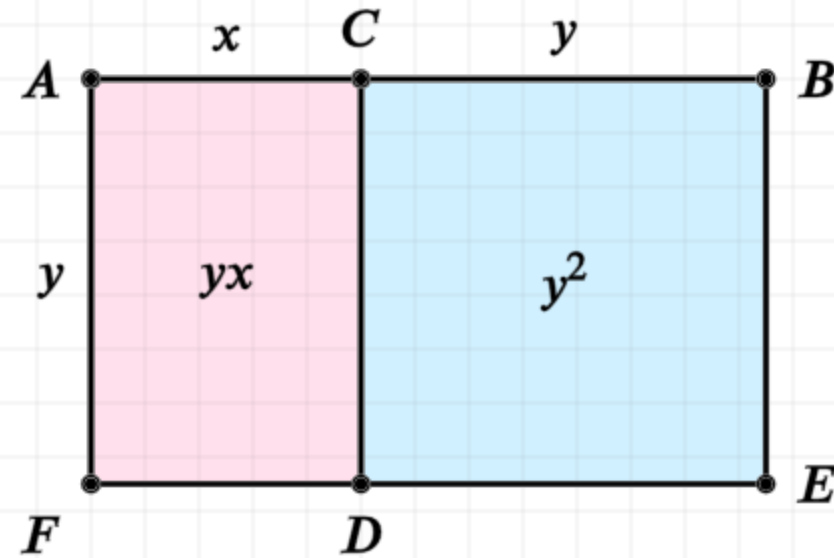
Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

$$AB = AC + CB$$

$$CB = BE = CD = AF$$

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

The rectangle AE is the sum of the rectangles AF and CE

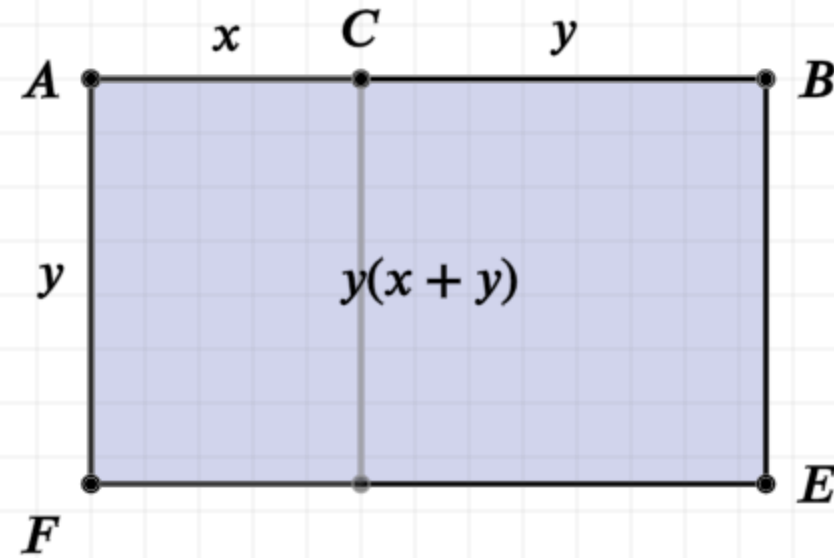
$$AB = AC + CB$$

$$CB = BE = CD = AF$$

$$\text{blue } AE = \text{pink } AD + \text{light blue } CE$$

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



$$AB = AC + CB$$

$$CB = BE = CD = AF$$

$$AE = AD + CE$$

$$AE = AF \cdot AB, \quad \therefore AE = CB \cdot AB$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

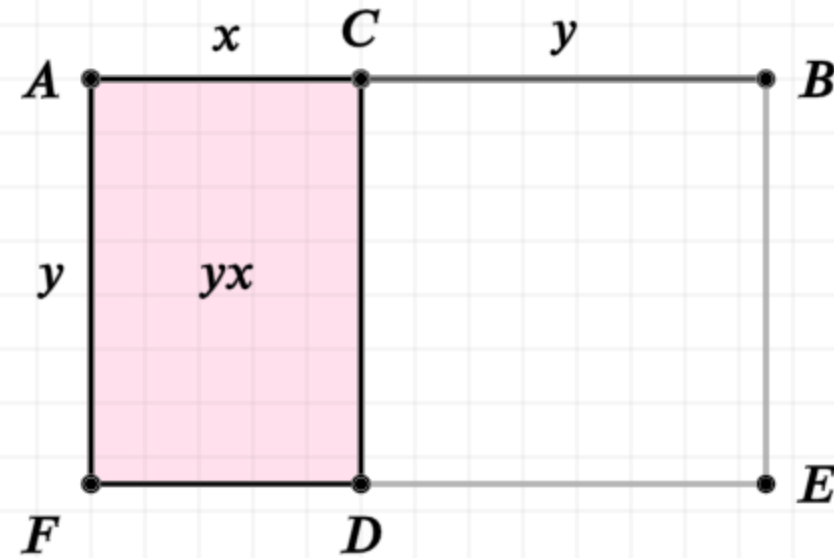
Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

The rectangle AE is the sum of the rectangles AF and CE

Since AF is equal in length to CB (I.34), the rectangle AE is equal to the rectangle contained by lines AB and CB

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

The rectangle AE is the sum of the rectangles AF and CE

Since AF is equal in length to CB (I.34), the rectangle AE is equal to the rectangle contained by lines AB and CB

Similarly, the rectangle AD is equal to the rectangle contained by lines CB and AC

$$AB = AC + CB$$

$$CB = BE = CD = AF$$

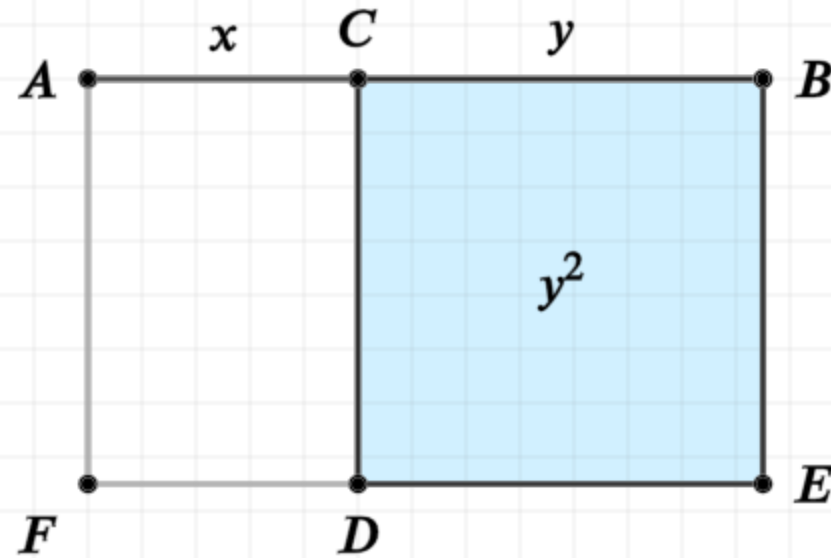
$$AE = AD + CE$$

$$AE = AF \cdot AB, \quad \therefore AE = CB \cdot AB$$

$$AD = AF \cdot AC, \quad \therefore AD = CB \cdot AC$$

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



$$AB = AC + CB$$

$$CB = BE = CD = AF$$

$$AE = AD + CE$$

$$AE = AF \cdot AB, \quad \therefore AE = CB \cdot AB$$

$$AD = AF \cdot AC, \quad \therefore AD = CB \cdot AC$$

$$CE = CD \cdot CB, \quad \therefore CE = CB^2$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

The rectangle AE is the sum of the rectangles AF and CE

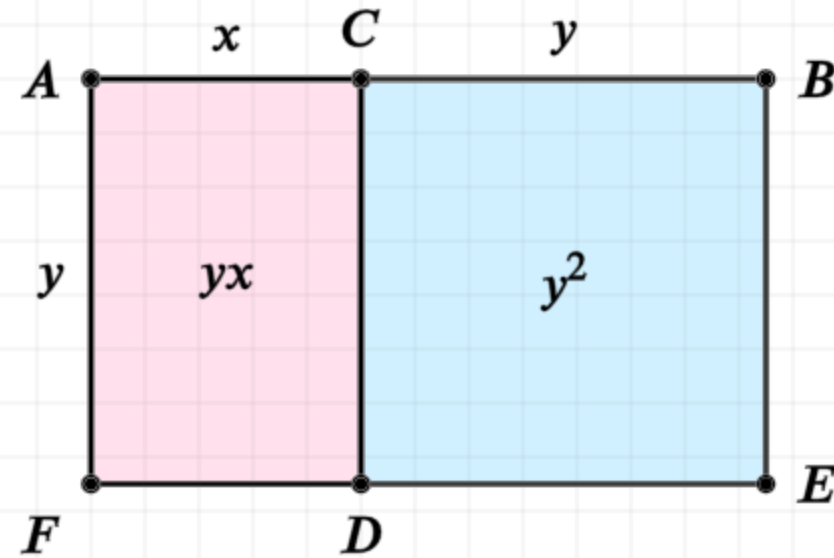
Since AF is equal in length to CB (I.34), the rectangle AE is equal to the rectangle contained by lines AB and CB

Similarly, the rectangle AD is equal to the rectangle contained by lines CB and AC

Since CD equals AF (I.34), CE is equal to the rectangle contained by lines CB and CB

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



$$AB = AC + CB$$

$$CB = BE = CD = AF$$

$$\text{Blue } AE = \text{Pink } AD + \text{Blue } CE$$

$$\text{Blue } AE = AF \cdot AB, \quad \therefore \text{Blue } AE = CB \cdot AB$$

$$\text{Pink } AD = AF \cdot AC, \quad \therefore \text{Pink } AD = CB \cdot AC$$

$$\text{Blue } CE = CD \cdot CB, \quad \therefore \text{Blue } CE = CB^2$$

$$CB \cdot AB = CB \cdot AC + CB^2$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

The rectangle AE is the sum of the rectangles AF and CE

Since AF is equal in length to CB (I.34), the rectangle AE is equal to the rectangle contained by lines AB and CB

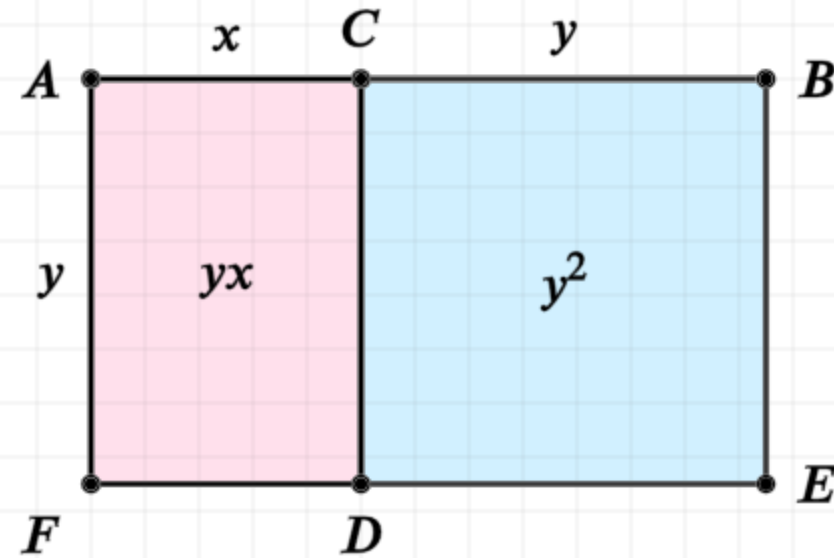
Similarly, the rectangle AD is equal to the rectangle contained by lines CB and AC

Since CD equals AF (I.34), CE is equal to the rectangle contained by lines CB and CB

Thus, the rectangle formed by CB,AB is equal to the sum of the rectangles formed by CB,AC and CB,CB

Proposition 3 of Book 2

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$CB \cdot AB = CB \cdot AC + CB^2$$

Proof:

Draw a square CDEB on the line CB (I.46) and draw a line AF parallel to either CD or BE (I.31). Extend DE to the point F

The rectangle AE is the sum of the rectangles AF and CE

Since AF is equal in length to CB (I.34), the rectangle AE is equal to the rectangle contained by lines AB and CB

Similarly, the rectangle AD is equal to the rectangle contained by lines CB and AC

Since CD equals AF (I.34), CE is equal to the rectangle contained by lines CB and CB

Thus, the rectangle formed by CB,AB is equal to the sum of the rectangles formed by CB,AC and CB,CB

$$AB = AC + CB$$

$$CB = BE = CD = AF$$

$$AE = AD + CE$$

$$AE = AF \cdot AB, \quad \therefore AE = CB \cdot AB$$

$$AD = AF \cdot AC, \quad \therefore AD = CB \cdot AC$$

$$CE = CD \cdot CB, \quad \therefore CE = CB^2$$

$$CB \cdot AB = CB \cdot AC + CB^2, \quad y(x + y) = yx + y^2$$

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